

SOFTWARE ENGINEERING

LAB 2: Agile backlog creation and sprint simulation in Jira

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SCREENSHOT 1: Backlog with Epics and User Stories

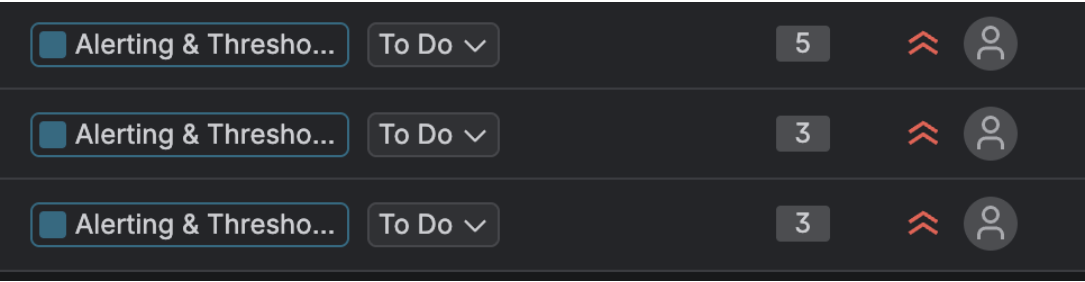
The screenshot displays the Jira interface for a project named 'Database Query Performance Profiler'. The left sidebar shows navigation options like 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', 'Filters', 'Dashboards', 'Teams', 'Goals', and 'Projects'. The main area shows the 'Backlog' view with a search bar and a 'Filter' button. Below the search bar, there's a section for 'Epic' with a list of items: 'No epic', 'Query Ingestion & Real-Time Monitoring', 'Execution Plan Analysis & Index Recommendations', and 'Alerting & Threshold Management'. The 'Backlog' section shows a list of 6 work items, including 'SCRUM-9' through 'SCRUM-14', each with a status (e.g., 'Query In...', 'Execution...'), a priority (e.g., 'To Do'), and a point value (e.g., 5, 3, 8). A '+ Create' button is visible at the bottom left.

SCREENSHOT 2: Story point

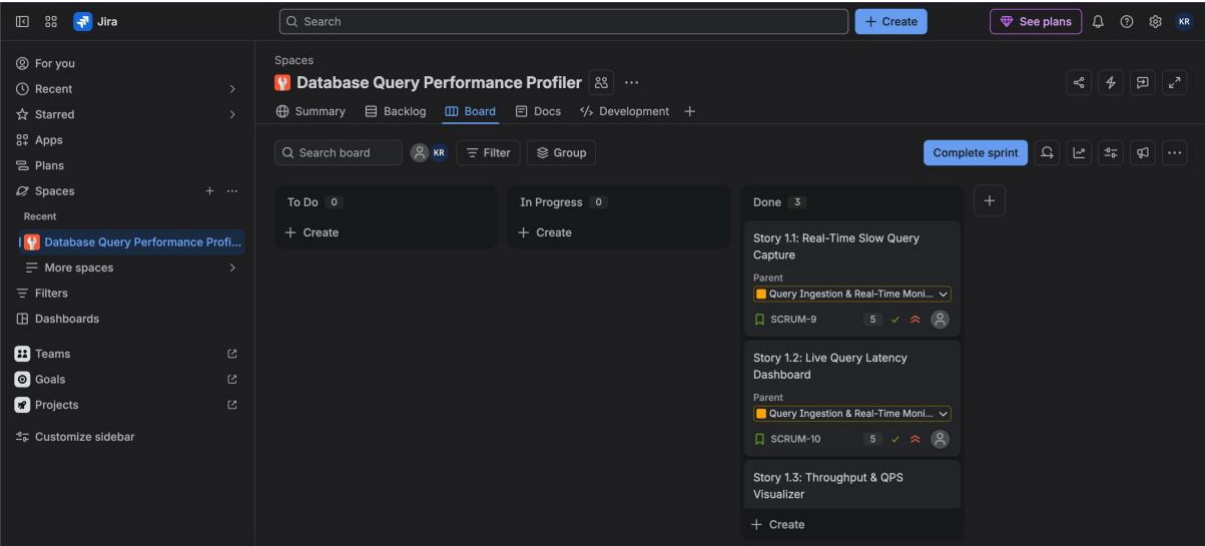
This close-up view of the Jira backlog items shows the following details:

Item	Status	Priority	Points	Assignee
Query In...	To Do	5	5	[User Icon]
Query In...	To Do	5	5	[User Icon]
Query In...	To Do	3	3	[User Icon]
Execution...	To Do	8	8	[User Icon]
Execution...	To Do	5	5	[User Icon]
Execution...	To Do	3	3	[User Icon]

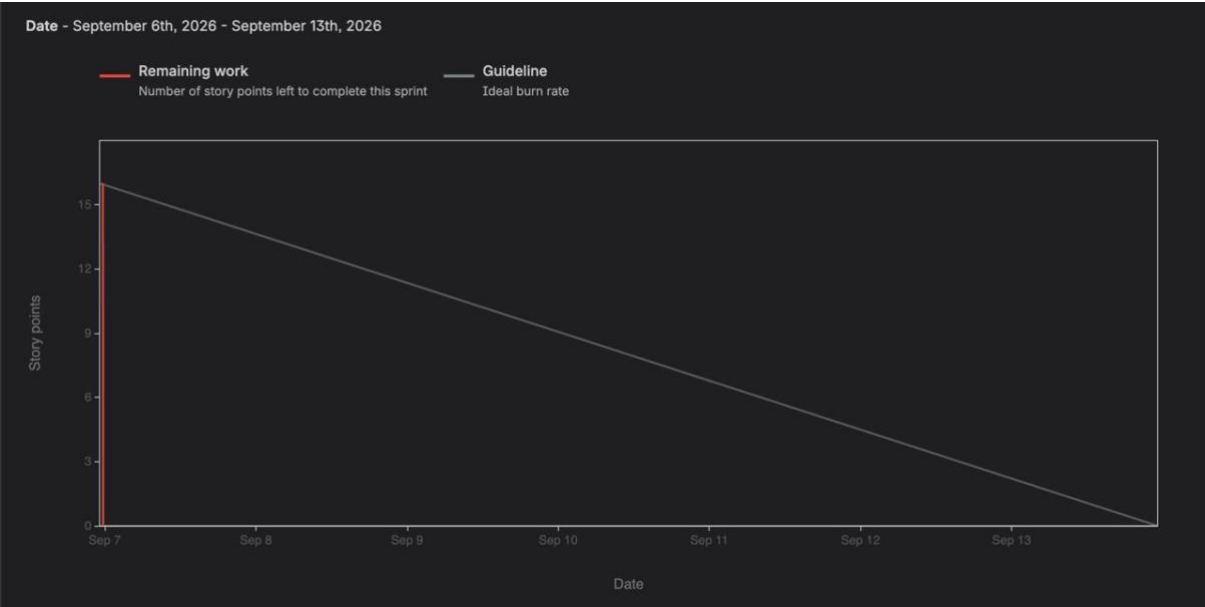
The word 'assignments' is visible at the bottom right of the image.



SCREENSHOT 3:



SCREENSHOT 4:



Answers to Reflection Questions

Q1. Did your estimations reflect the actual effort?

Answer: In our Planning Poker estimation, user stories involving complex algorithm visualizers (such as Story 2.1: *Query Execution Plan Visualizer*) were estimated at 8 story points due to the complexity of parsing raw execution trees and joins. During the simulated sprint, stories sized at 3 and 5 story points (such as slow query logging and latency dashboarding) proved significantly more modular and predictable. This reinforces that larger stories exceeding 5 points carry higher uncertainty and should ideally be decomposed into smaller sub-tasks.

Q2. Was your backlog well-prioritized?

Answer: Yes, the backlog was prioritized strictly by dependency and business value. Fundamental metric collection (ingesting slow queries and logging latency) was prioritized as High in Sprint 1 because advanced optimization features in Sprint 2 (such as index recommendation algorithms and plan visualizers) rely entirely on having reliable incoming query data.

Q3. How did your simulated sprint align with your plan?

Answer: The simulated sprint closely aligned with our planned sprint scope. Sprint 1 committed 13 story points and Sprint 2 committed 16 story points. All prioritized user stories successfully transitioned from To Do → In Progress → Done, meeting the Sprint Goal of delivering real-time performance tracking and index advisory capabilities within the 1-week timebox.

Q4. What insights did the burndown chart give about your team's capacity?

Answer: The burndown chart illustrated the total remaining effort dropping to zero as all tasks were completed. The sharp drop reflects completion of tasks near the end of the sprint cycle, demonstrating that batch verification can create sudden dips compared to the steady linear guideline. It indicates our team has a sustainable sprint capacity of approximately 13 to 16 story points per 1-week sprint.